

SUMMARY

- AI master’s student at École Polytechnique; research experience at Inria WILLOW; co-author of **KDD 2025 (oral)** and **ICCV 2025** papers on multimodal video understanding.
- **3.5 years of software engineering experience** building production systems; trained and evaluated models across robot learning, LLM post-training with RL, and diffusion distillation.
- Research interests: robot learning, multimodality, representation learning, diffusion/flow matching, and LLM/VLM pre- and post-training.

EDUCATION

École Polytechnique	Sept 2025 – Present
Masters of Science and Technology in Trustworthy and Responsible AI	Paris, France
Birla Institute of Technology And Science (BITS), Pilani	Aug 2017 – Aug 2021
Bachelors of Engineering in Computer Science	Pilani, India

SELECTED PUBLICATIONS

- VideoConviction: A Multimodal Benchmark for Human Conviction and Stock Market Recommendations – **KDD’25 Oral**
- FinCap: Topic-Aligned Captions for Short-Form Financial YouTube Videos – **ICCV’25 Poster**

WORK EXPERIENCE

Inria Paris, WILLOW Team Research Intern	Paris, France Apr 2026 – Present
Advised by Cordelia Schmid and Shizhe Chen	

- **Sim Data Engine & 3D Pretraining:** Built a MuJoCo data engine (188K point clouds, 4.2M+ labels) and pretrained a Point Transformer V3 encoder: increased success rate on RLBench tasks by 20%.
- **Language-Grounded VLA Backbone:** Extended the encoder to joint 3D segmentation and language grounding, powering 3 VLAs (PointACT, QwenGR00T, EO1) on RLBench and RoboCasa.

Georgia Institute of Technology Research Intern	Atlanta, Georgia (Remote) May 2024 – Sep 2025
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- **VideoConviction:** Built a multimodal benchmark (288 videos, 6K+ annotations, 457 hours) to study conviction cues in financial advice, benchmarking transformer-based MLLMs (Gemini 2.5, GPT-4o).
- **FinCap:** Developed a short-form financial video captioning benchmark (624 clips, 7 modality setups, 5 tasks), benchmarking modality fusion strategies for recommendation, visual features, sentiment, and entity extraction.

Urban Company Software Developer 2 (Promoted in 2023)	Bangalore, India Aug 2021 – Mar 2025
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- **Product Catalog Cache:** Designed a distributed cache datastore for 27K products across 5 countries using Kafka, Redis, and MongoDB; reduced service latency by 40% and supported recommendation pipelines.
- **Dynamic Pricing Framework:** Deployed SARIMA + XGBoost-based time-series models for adaptive pricing by demand and region, improving revenue by 4% on \$80M+ transactions.

IIIT Delhi Research Intern	New Delhi, India Jan 2021 – June 2021
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- **Author Profiling:** Modeled authorship patterns in scientific publications using GNNs (GNN-XML) and AnnexML (KNN) on the S2ORC corpus (600K papers, 160K authors), achieving P@1: 15%, P@3: 27%.

SELECTED PROJECTS

- **RL Post-Training for LLM Agents:** Trained Qwen3-1.7B on ALFWorld household tasks with SFT + RLOO built from scratch; RL raised success on 134 unseen tasks from **58.2% to 64.6%** ($p = 0.019$).
- **Planner Atlas:** CEM planning in a JEPA world model (PushT) picks plans with **2.6×** the model error of random ones; fine-tuning on them cut plan-selection regret **14%** vs. random data (12 seeds, $p=0.0015$).
- **Fast Diffusion via LoRA Distillation:** Distilled SD1.5 into a LoRA student for 2–8 step text-to-image generation (up to 18× faster); built a CUDA-timed benchmark with CLIP alignment + qualitative grids.

KEY ACCOMPLISHMENTS AND EXPERIENCES

- **Charpak Masters Scholarship:** 56 selected/2500+ applicants – French Ministry of Europe & Foreign Affairs.
- **Competitive Programming:** World rank 4, 6 in codechef long challenge Sept’22, Oct’22 (10K+ participants).

SKILLS

Languages	Python, C++, JavaScript, C
ML Frameworks	PyTorch, Hugging Face (Transformers, Diffusers, PEFT), NumPy
ML / Research	LLM Post-Training, Reinforcement Learning, Robot Learning, Multimodal Learning, Diffusion
Systems / Tools	Linux, Git, Docker, SLURM, MuJoCo, Kafka, Redis, MongoDB, AWS